In [43]:
```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
pio.renderers.default = "notebook"
import plotly.colors as colors
pio.templates.default = "plotly_white"
```

In [9]:
```
data = pd.read_csv("Sample - Superstore.csv",encoding = 'latin-1')  ##encoding='
```

In [10]:
```
data
```

Out[10]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segme
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consun
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consun
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corpor
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consun
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consun
...	...	...	...	...	...	...	...	
9989	9990	CA-2014-110422	1/21/2014	1/23/2014	Second Class	TB-21400	Tom Boeckenhauer	Consun
9990	9991	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consun
9991	9992	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consun
9992	9993	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consun
9993	9994	CA-2017-119914	5/4/2017	5/9/2017	Second Class	CC-12220	Chris Cortes	Consun

9994 rows × 21 columns

In [22]:
```
data.head()
```

Out[22]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 21 columns

Let’s start by looking at the descriptive statistics of the dataset

In [12]:
```
data.describe()
```

Out[12]:

	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000

In [13]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   str
 2   Order Date     9994 non-null   str
 3   Ship Date      9994 non-null   str
 4   Ship Mode      9994 non-null   str
 5   Customer ID    9994 non-null   str
 6   Customer Name  9994 non-null   str
 7   Segment        9994 non-null   str
 8   Country        9994 non-null   str
 9   City           9994 non-null   str
 10  State          9994 non-null   str
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   str
 13  Product ID     9994 non-null   str
 14  Category       9994 non-null   str
 15  Sub-Category   9994 non-null   str
 16  Product Name   9994 non-null   str
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), str(15)
memory usage: 1.6 MB
```

Converting data columns

In [18]:
```
data['Order Date'] = pd.to_datetime(data['Order Date'])
```

In [19]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   str
 2   Order Date     9994 non-null   datetime64[us]
 3   Ship Date      9994 non-null   str
 4   Ship Mode      9994 non-null   str
 5   Customer ID    9994 non-null   str
 6   Customer Name  9994 non-null   str
 7   Segment        9994 non-null   str
 8   Country        9994 non-null   str
 9   City           9994 non-null   str
 10  State          9994 non-null   str
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   str
 13  Product ID     9994 non-null   str
 14  Category       9994 non-null   str
 15  Sub-Category   9994 non-null   str
 16  Product Name   9994 non-null   str
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: datetime64[us](1), float64(3), int64(3), str(14)
memory usage: 1.6 MB
```

In [20]:
```
data['Ship Date'] = pd.to_datetime(data['Ship Date'])
```

In [21]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   str
 2   Order Date     9994 non-null   datetime64[us]
 3   Ship Date      9994 non-null   datetime64[us]
 4   Ship Mode      9994 non-null   str
 5   Customer ID    9994 non-null   str
 6   Customer Name  9994 non-null   str
 7   Segment        9994 non-null   str
 8   Country        9994 non-null   str
 9   City           9994 non-null   str
 10  State          9994 non-null   str
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   str
 13  Product ID     9994 non-null   str
 14  Category       9994 non-null   str
 15  Sub-Category   9994 non-null   str
 16  Product Name   9994 non-null   str
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: datetime64[us](2), float64(3), int64(3), str(13)
memory usage: 1.6 MB
```

In [23]:
```
data.head()
```

Out[23]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 21 columns

Adding New Date-Based Columns

In [25]:
```
data['Order Month'] = data['Order Date'].dt.month
data['Order Year'] = data['Order Date'].dt.year
data['Order Day of Week'] = data['Order Date'].dt.dayofweek
```

In [26]:
```
data.head()
```

Out[26]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 24 columns

Monthly sales analysis

In [35]:
```
sales_by_month = data.groupby('Order Month')['Sales'].sum().reset_index()
```

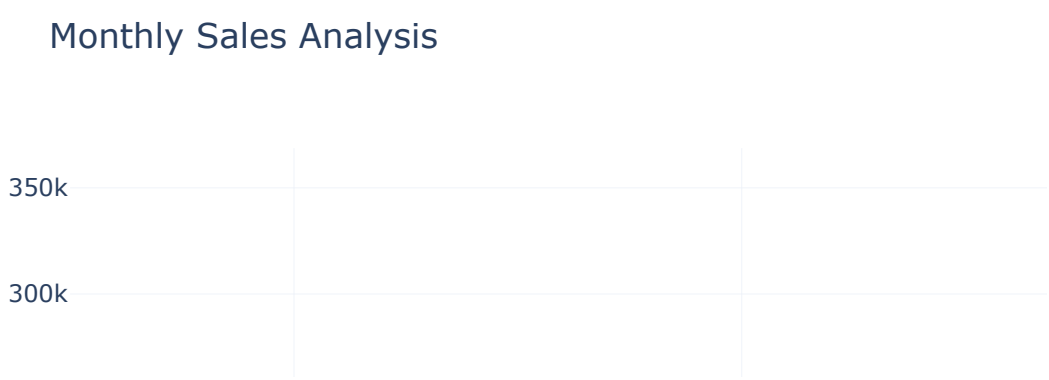
In [36]:
```
sales_by_month
```

Out[36]:

	Order Month	Sales
0	1	94924.8356
1	2	59751.2514
2	3	205005.4888
3	4	137762.1286
4	5	155028.8117
5	6	152718.6793
6	7	147238.0970
7	8	159044.0630
8	9	307649.9457
9	10	200322.9847
10	11	352461.0710
11	12	325293.5035

In [44]:

```
fig = px.line(sales_by_month,
              x='Order Month',
              y='Sales',
              title='Monthly Sales Analysis')
fig.show()
```



In [45]:
```
data.head()
```

Out[45]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 24 columns

Sales analysis by Category

In [46]:
```
sales_by_category = data.groupby(['Category'])['Sales'].sum().reset_index()
```

In [47]:
```
sales_by_category
```

Out[47]:

	Category	Sales
0	Furniture	741999.7953
1	Office Supplies	719047.0320
2	Technology	836154.0330

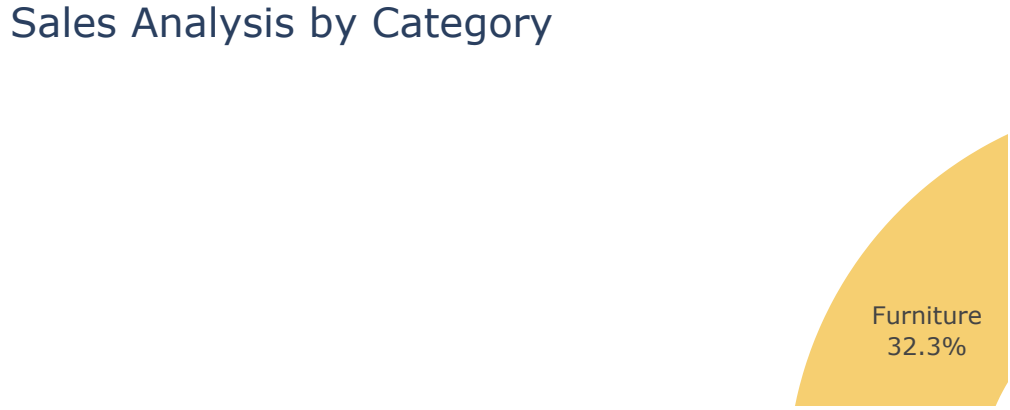
In [81]:
```
#sales_by_category = data.groupby('Category')['Sales'].sum().reset_index()


fig = px.pie(sales_by_category,
             values='Sales',
             names='Category',
             hole=0.5,
```

```
            color_discrete_sequence=px.colors.qualitative.Pastel)

fig.update_traces(textposition='inside', textinfo='percent+label')
fig.update_layout(title_text='Sales Analysis by Category', title_font=dict(size=

fig.show()
```



Sales analysis by Sub Category

In [49]:
```
data.head()
```

Out[49]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 24 columns

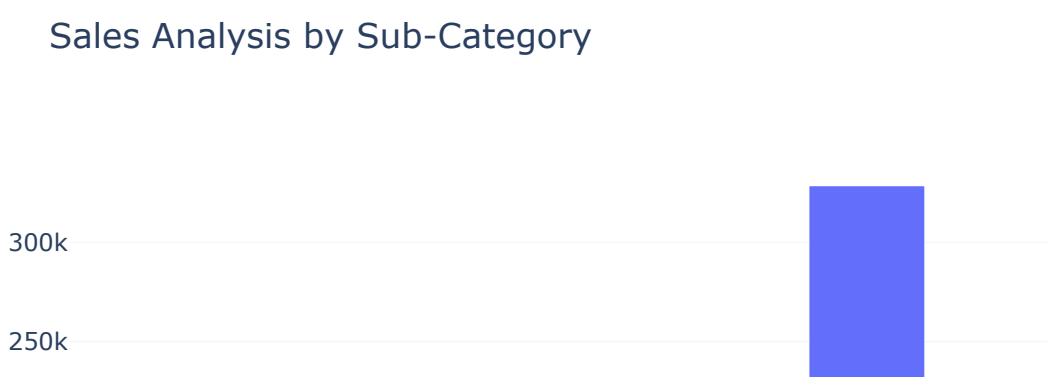
In [54]:
```
sales_by_subcategory = data.groupby('Sub-Category')['Sales'].sum().reset_index()
```

In [55]:
```
sales_by_subcategory
```

Out[55]:

	Sub-Category	Sales
0	Accessories	167380.3180
1	Appliances	107532.1610
2	Art	27118.7920
3	Binders	203412.7330
4	Bookcases	114879.9963
5	Chairs	328449.1030
6	Copiers	149528.0300
7	Envelopes	16476.4020
8	Fasteners	3024.2800
9	Furnishings	91705.1640
10	Labels	12486.3120
11	Machines	189238.6310
12	Paper	78479.2060
13	Phones	330007.0540
14	Storage	223843.6080
15	Supplies	46673.5380
16	Tables	206965.5320

In [56]:
```
#sales_by_subcategory = data.groupby('Sub-Category')['Sales'].sum().reset_index(
fig = px.bar(sales_by_subcategory,
             x='Sub-Category',
             y='Sales',
             title='Sales Analysis by Sub-Category')
fig.show()
```



Monthly profit analysis

In [57]:
```
data.head()
```

Out[57]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 24 columns

In [60]:
```
profit_by_month = data.groupby('Order Month')['Profit'].sum().reset_index()
```

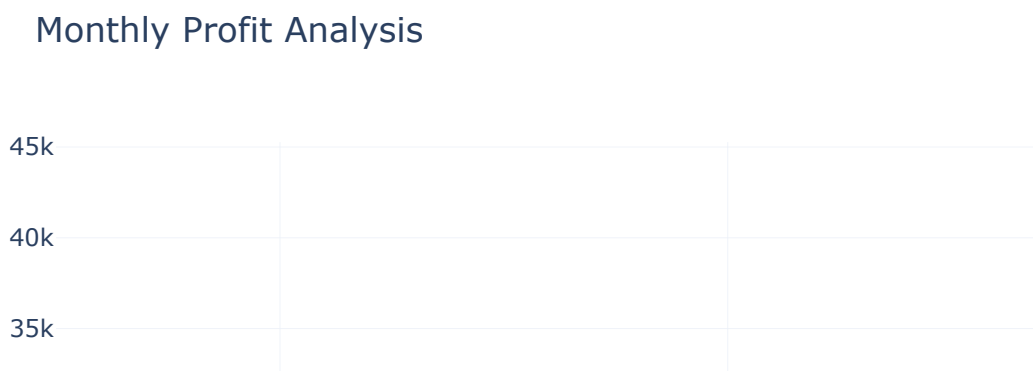
In [61]:
```
profit_by_month
```

Out[61]:

	Order Month	Profit
0	1	9134.4461
1	2	10294.6107
2	3	28594.6872
3	4	11587.4363
4	5	22411.3078
5	6	21285.7954
6	7	13832.6648
7	8	21776.9384
8	9	36857.4753
9	10	31784.0413
10	11	35468.4265
11	12	43369.1919

In [65]:

```
fig = px.line(profit_by_month,
              x='Order Month',
              y='Profit',
              title='Monthly Profit Analysis')
fig.show()
```



Profit analysis by category

In [67]:
```
profit_by_Category = data.groupby('Category')['Profit'].sum().reset_index()
```

In [68]:
```
profit_by_Category
```

Out[68]:

	Category	Profit
0	Furniture	18451.2728
1	Office Supplies	122490.8008
2	Technology	145454.9481

In [80]:
```
fig = px.pie(profit_by_Category,
             values='Profit',
             names='Category',
             hole=0.5,
             color_discrete_sequence=px.colors.qualitative.Pastel)

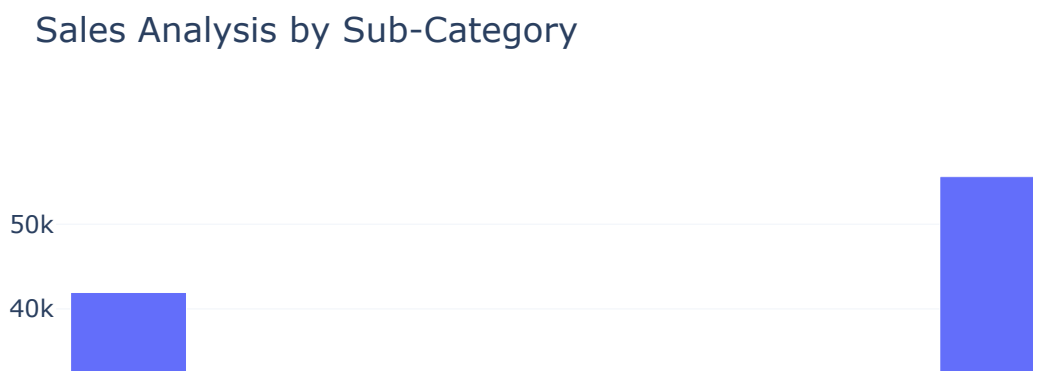
fig.update_traces(textposition='inside', textinfo='percent+label')
fig.update_layout(title_text='Profit Analysis by Category', title_font=dict(size

fig.show()
```

Profit Analysis by Category

Profit annalysis by Sub Category

In [77]:
```
Profit_by_subcategory = data.groupby('Sub-Category')['Profit'].sum().reset_index
fig = px.bar(Profit_by_subcategory,
             x='Sub-Category',
             y='Profit',
             title='Sales Analysis by Sub-Category')
fig.show()
```



Sales and Profit - customer Segment

In [78]:
```
sales_profit_by_segment = data.groupby('Segment').agg({'Sales': 'sum', 'Profit':

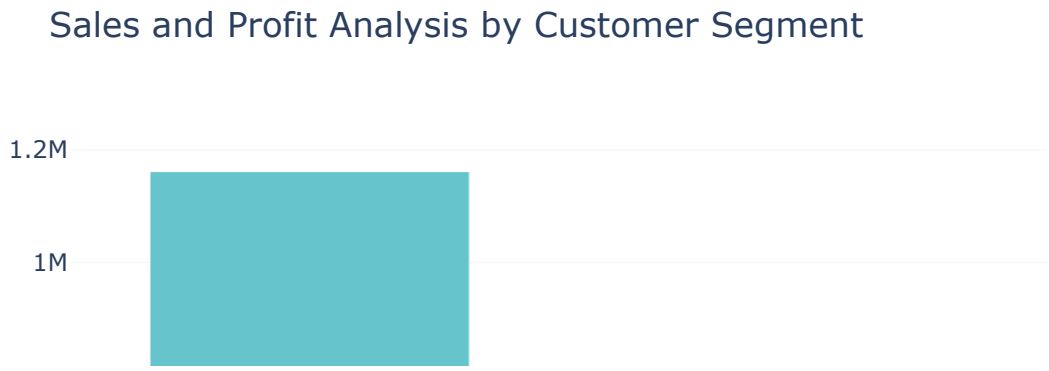
color_palette = colors.qualitative.Pastel

fig = go.Figure()
fig.add_trace(go.Bar(x=sales_profit_by_segment['Segment'],
                     y=sales_profit_by_segment['Sales'],
                     name='Sales',
                     marker_color=color_palette[0]))

fig.add_trace(go.Bar(x=sales_profit_by_segment['Segment'],
                     y=sales_profit_by_segment['Profit'],
                     name='Profit',
                     marker_color=color_palette[1]))

fig.update_layout(title='Sales and Profit Analysis by Customer Segment',
                  xaxis_title='Customer Segment', yaxis_title='Amount')

fig.show()
```



Analyse Sales to profit ratio

In [79]:
```
sales_profit_by_segment = data.groupby('Segment').agg({'Sales': 'sum', 'Profit':
sales_profit_by_segment['Sales_to_Profit_Ratio'] = sales_profit_by_segment['Sale
print(sales_profit_by_segment[['Segment', 'Sales_to_Profit_Ratio']])
```

```
       Segment  Sales_to_Profit_Ratio
0     Consumer               8.659471
1    Corporate               7.677245
2  Home Office               7.125416
```

In [ ]: